PROJECT REPORT ON

TET2-based Differential Gene Expression and Immune Profiling using TCGA Data

INTRODUCTION

Background on TCGA and RNA-seq:

A high-throughput method for assessing gene expression throughout the transcriptome is bulk RNA sequencing. It is essential to comprehending the biology of disease, particularly in the study of cancer. A comprehensive and openly accessible database, the Cancer Genome Atlas (TCGA) provides extensive genomic datasets, including RNA-seq and clinical metadata for different forms of cancer.

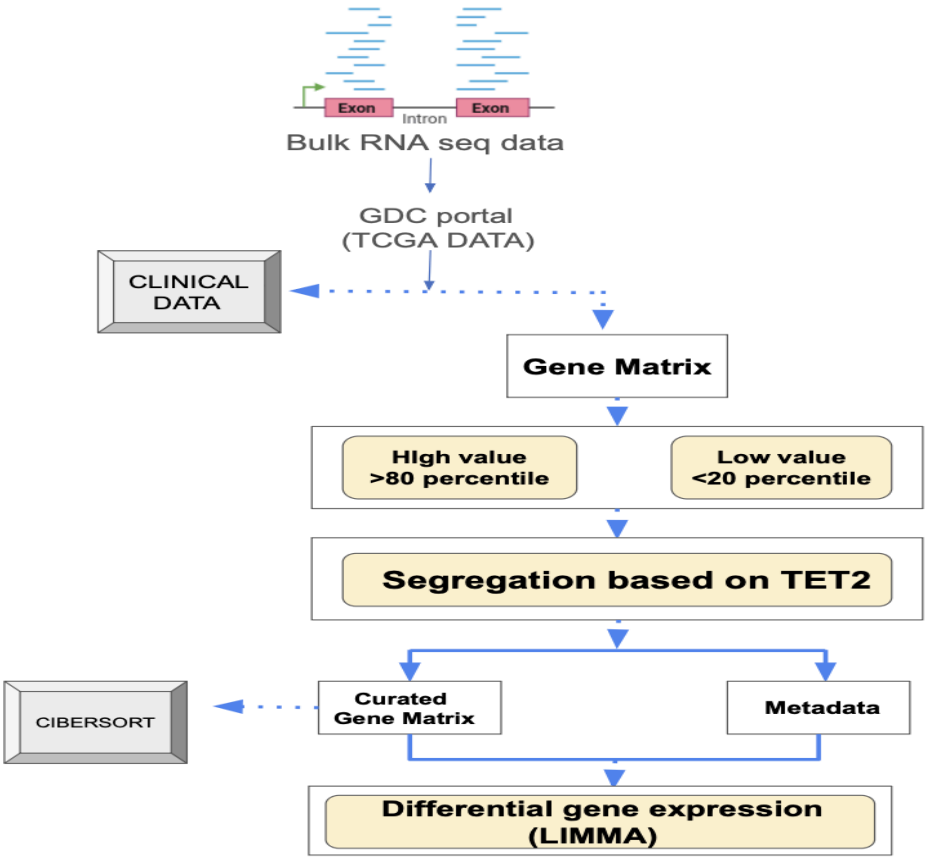
Pay attention to TET2:

The epigenetic regulator TET2 (Ten-Eleven Translocation 2) is known to regulate DNA demethylation. Its loss-of-function mutations or altered expression, linked to a number of malignancies, affect immune regulation and patient prognosis. Investigating the downstream transcriptional effects of TET2 expression levels and their relationship to immune cell infiltration is the main goal of this project.

OBJECTIVE

The main goal of this project was to use TCGA RNA-seq data to look into how TET2 expression affects the immune system in cancer. Based on the amount of TET2 present, patient samples were divided into two groups: TET2-high (above the 80th percentile) and TET2-low (below the 20th percentile). This was followed by immune cell deconvolution using the CIBERSORT method to guess the numbers of different types of immune cells in each sample. Then, a differential gene expression study was done between the TET2-high and TET2-low groups to find genes that had changed a lot. Finally, the information from the tests was shown in the form of helpful graphs and figures. The goal was to find trends related to immune modulation and transcriptional differences linked to TET2 expression.

Workflow



Data Acquisition

The Genomic Data Commons (GDC) portal was used to get bulk RNA sequencing data and clinical details from The Cancer Genome Atlas (TCGA). Normalized gene expression profiles from a variety of patient samples for a certain type of cancer were in the collection.

Gene Matrix Preparation

A clear and useful gene expression matrix was created using expression values from several input files, one for each sample. Annotation information was first extracted from

File 1, notably Columns 2 and 3, which contain gene names and symbols. This data was put into an empty reference DataFrame to build the gene matrix's backbone.

To a second blank DataFrame, column 7 from each sample file was inserted. TPM are expression values in this column. We modified the heading of each file's expression data column to reflect its filename or basename, thereby categorizing the data by sample.

After removing column 7 from each sample file, the reference DataFrame with gene names and expression columns was created. A full gene expression matrix was created with rows of genes and columns of data. In the last step of formatting, all the values were numbers, and the matrix was set up for immune deconvolution and differential gene expression tests.

Stratification Based on TET2 Expression

To study the impact of TET2 on gene expression and immune infiltration, samples were stratified based on TET2 expression levels. Using the 80th percentile as the threshold for high expression and the 20th percentile for low expression, two groups of samples were created: **TET2-high** and **TET2-low**. This segregation allowed for a clear comparison of molecular and immune features influenced by TET2 variation.

Immune Cell Profiling via CIBERSORT

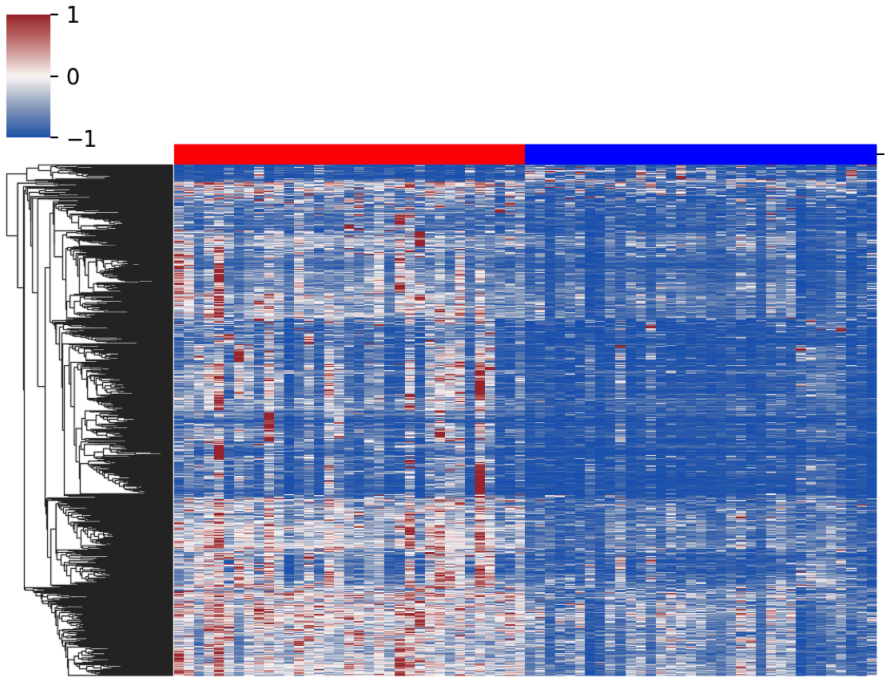
The curated gene expression matrix was processed using the CIBERSORT algorithm to perform immune cell deconvolution. CIBERSORT applies support vector regression to estimate the proportions of 22 immune cell types within each tumor sample, based on a predefined signature matrix (LM22). The resulting immune profile for each sample provided insights into how TET2 expression might be associated with variations in immune cell infiltration in the tumor microenvironment.

Differential Gene Expression Analysis (LIMMA)

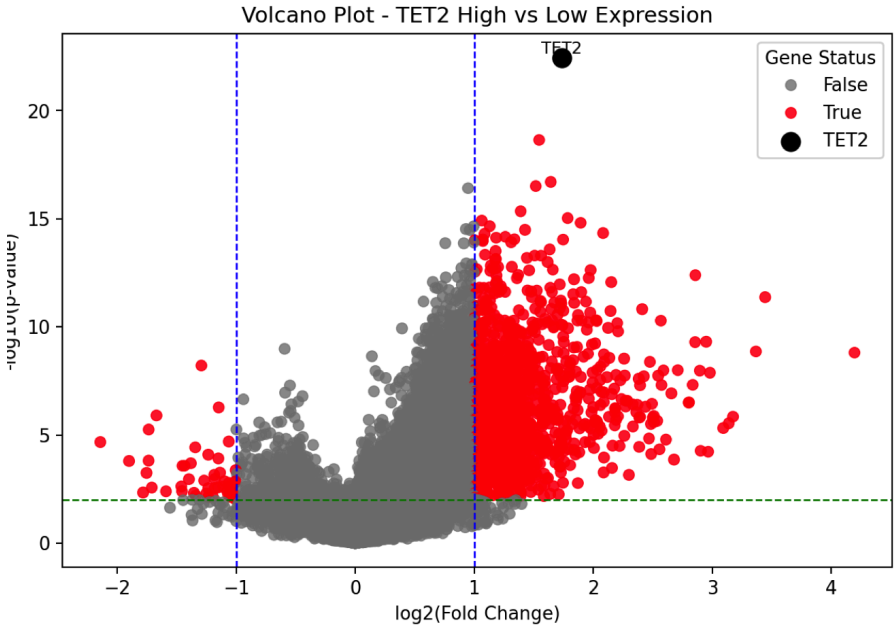
Following stratification, differential gene expression analysis was conducted between the TET2-high and TET2-low groups using the LIMMA (Linear Models for Microarray Data) framework. This statistical approach enabled the identification of genes significantly upregulated or downregulated in relation to TET2 expression. The analysis used a model fitting approach and adjusted for multiple testing using false discovery rate (FDR) correction to ensure statistical reliability.

DATA VISUALISATION

HEATMAP

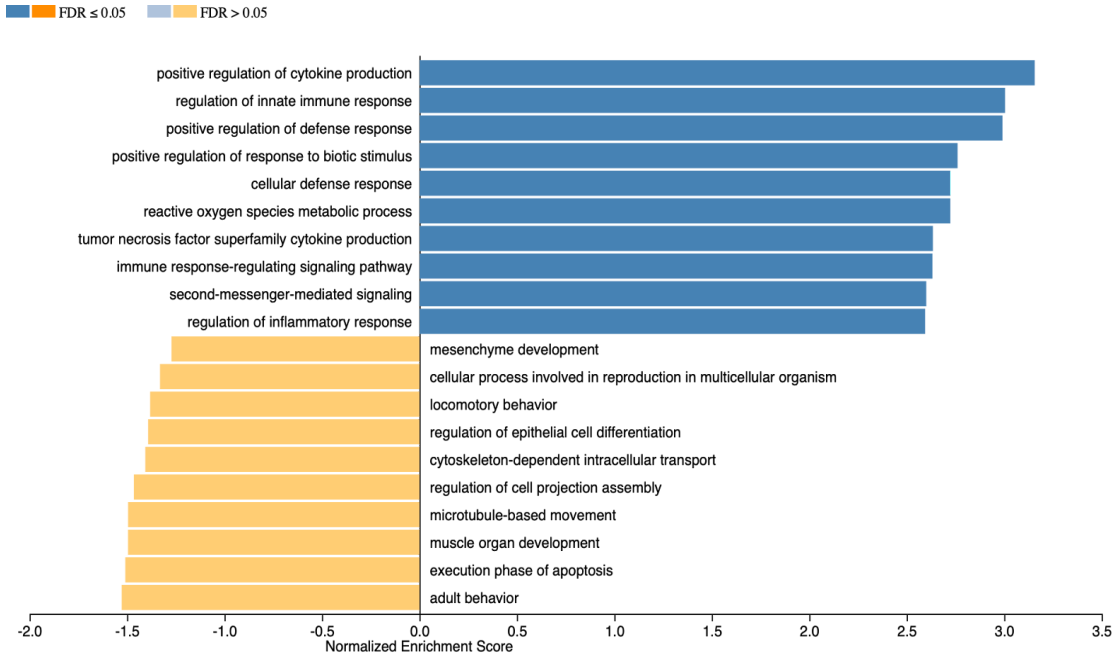


VOLCANO PLOT

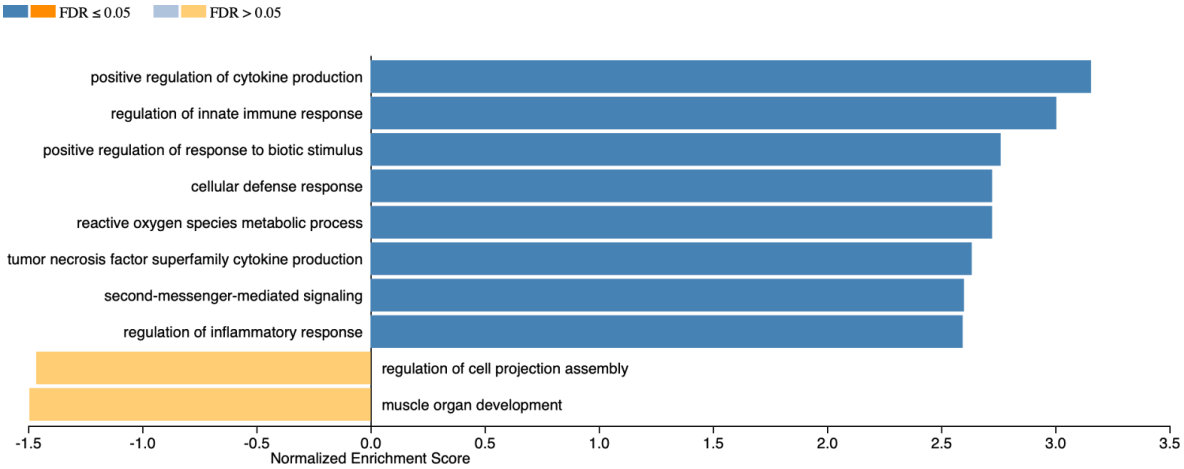


Pathway analysis:GSEA

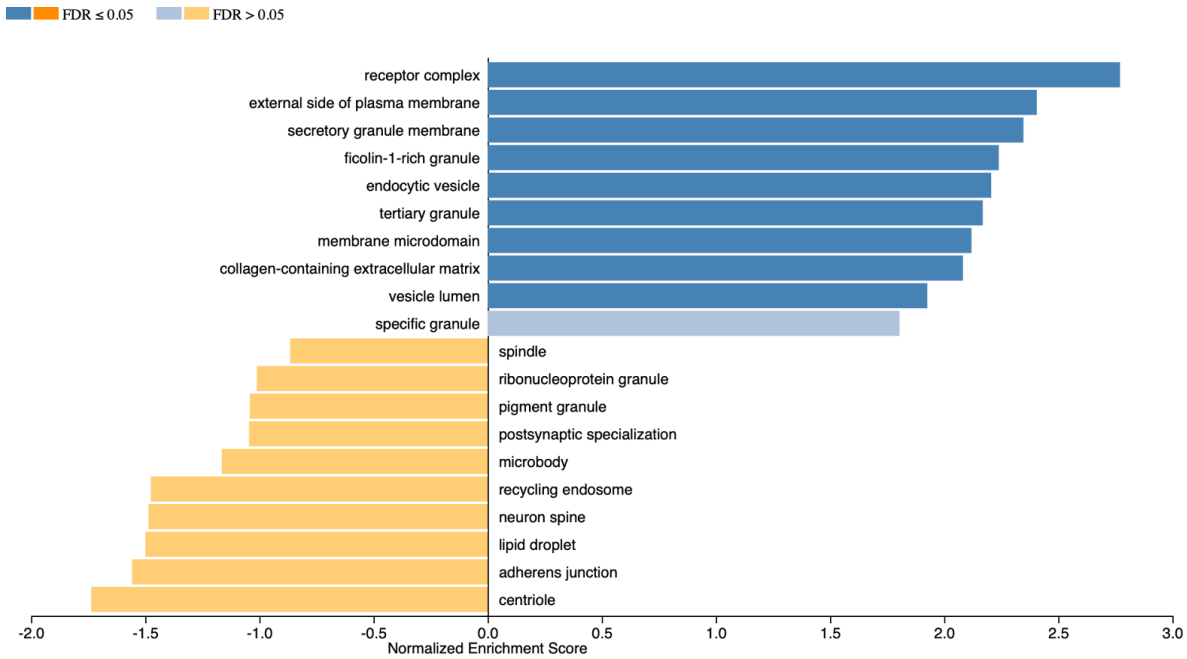
Enrichment Categories: geneontology_Biological_Process_noRedundant



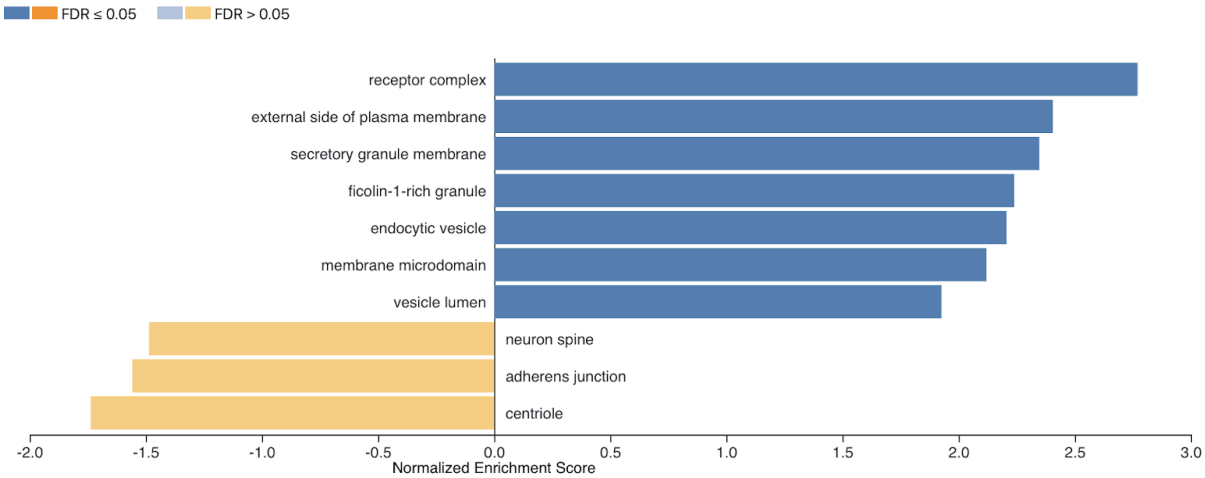
Weighted set cover



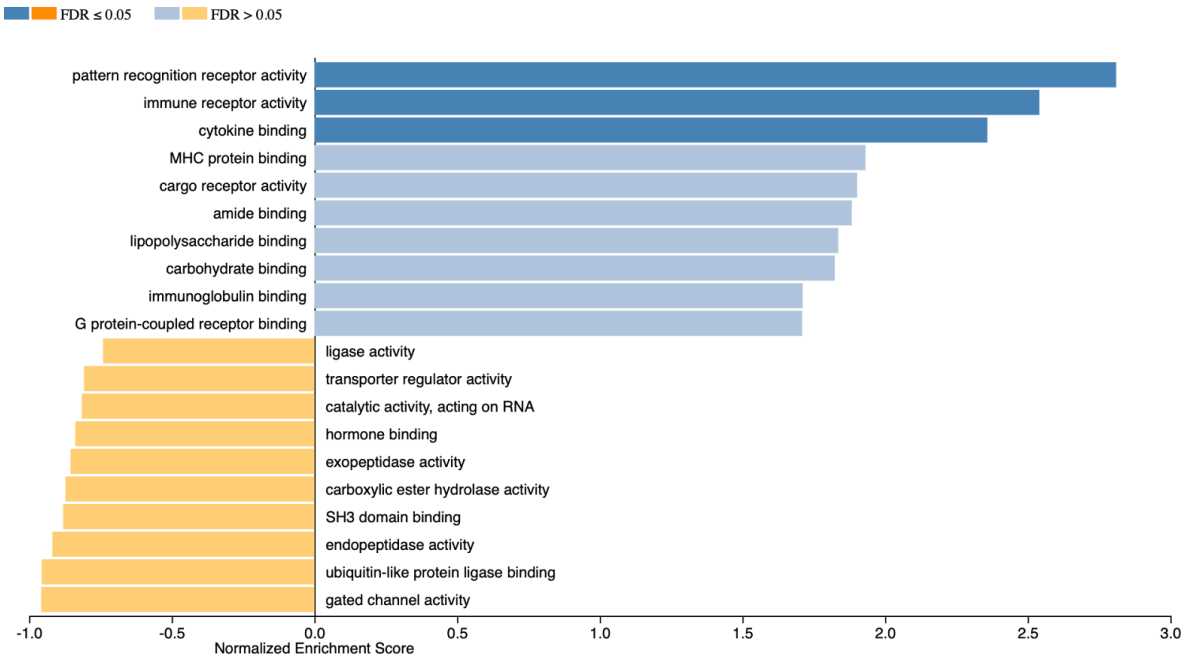
Enrichment Categories: geneontology_Cellular_Component_noRedundant



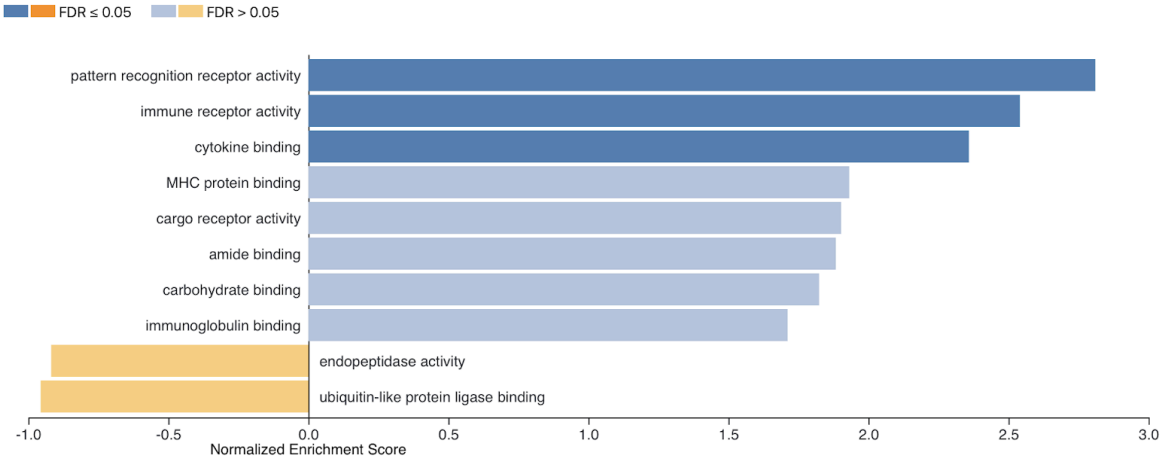
Weighted set cover



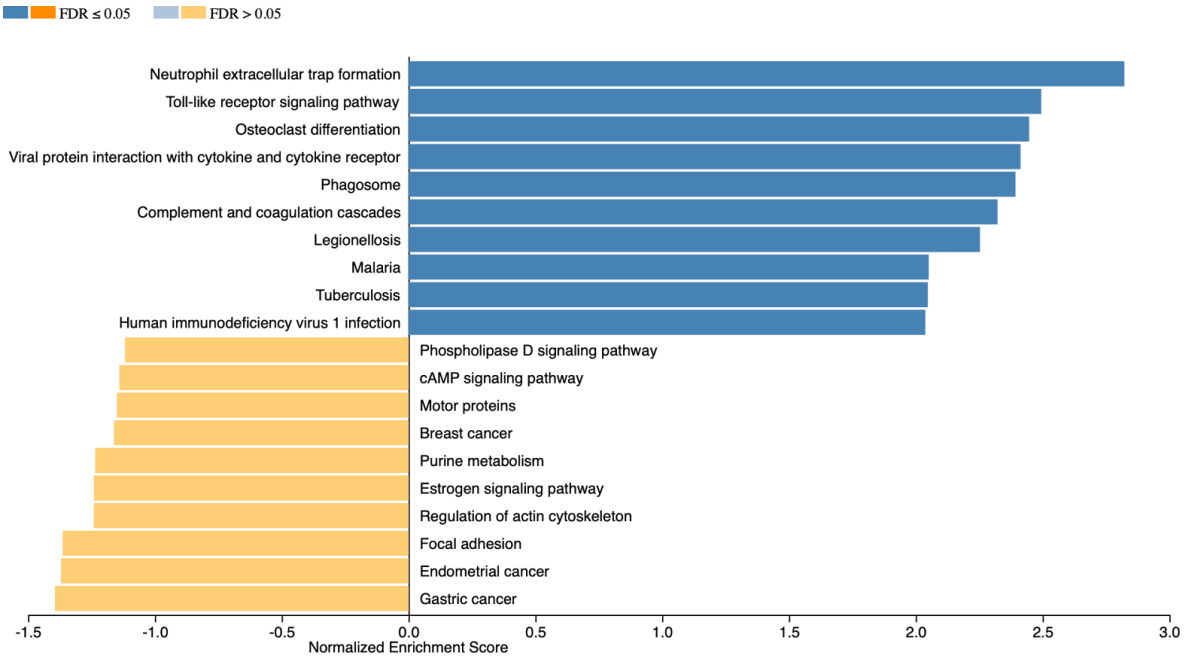
Enrichment Categories: geneontology_Molecular_Function_noRedundant



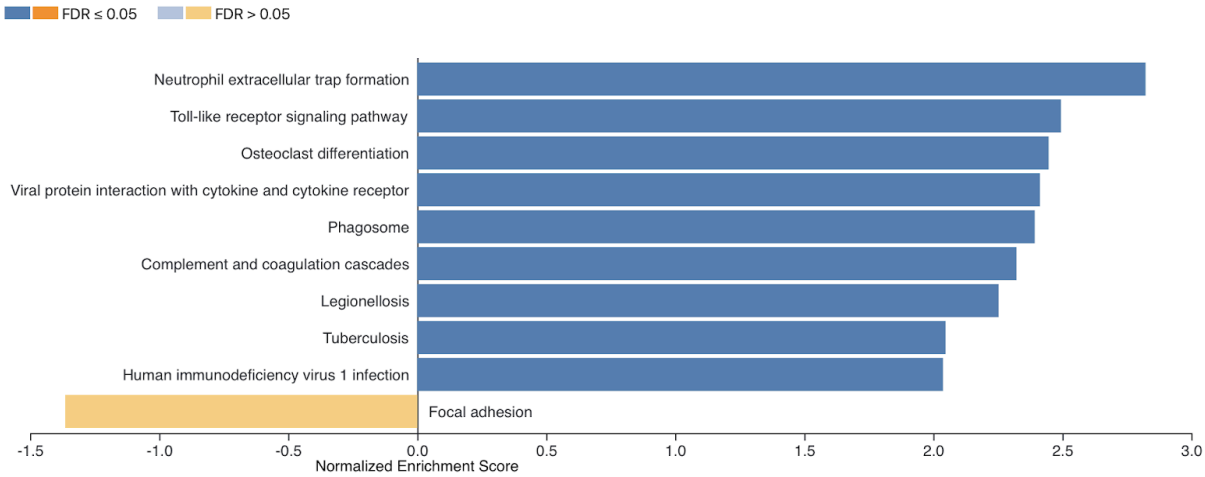
Weighted set cover



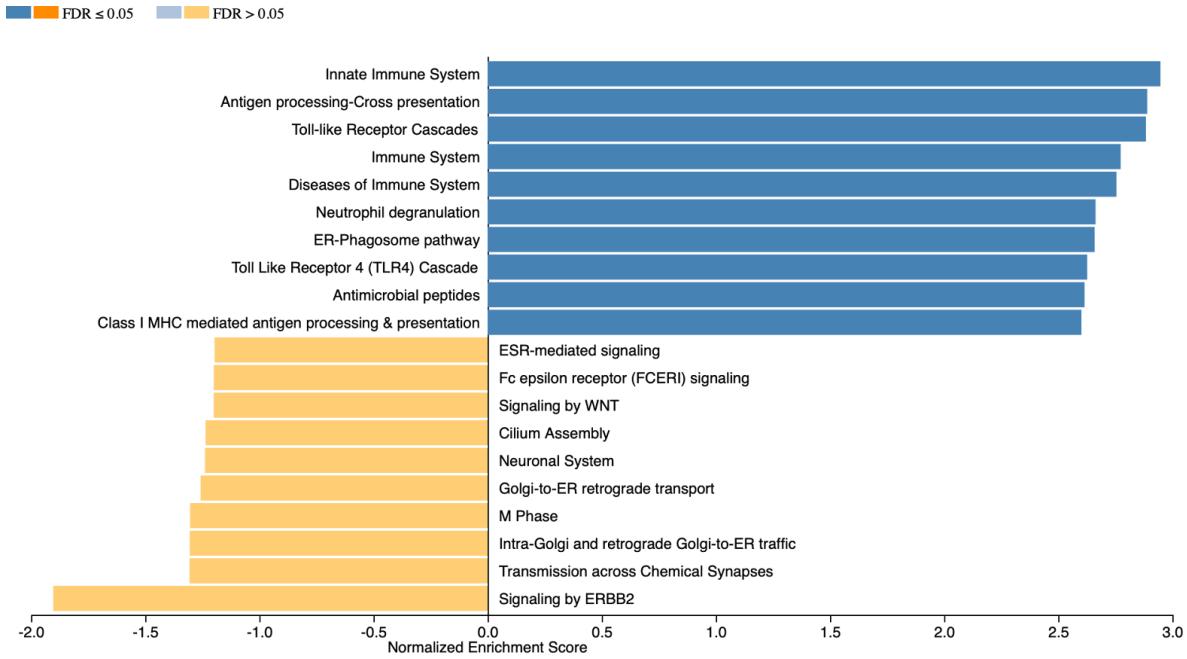
Enrichment Categories: pathway_KEGG



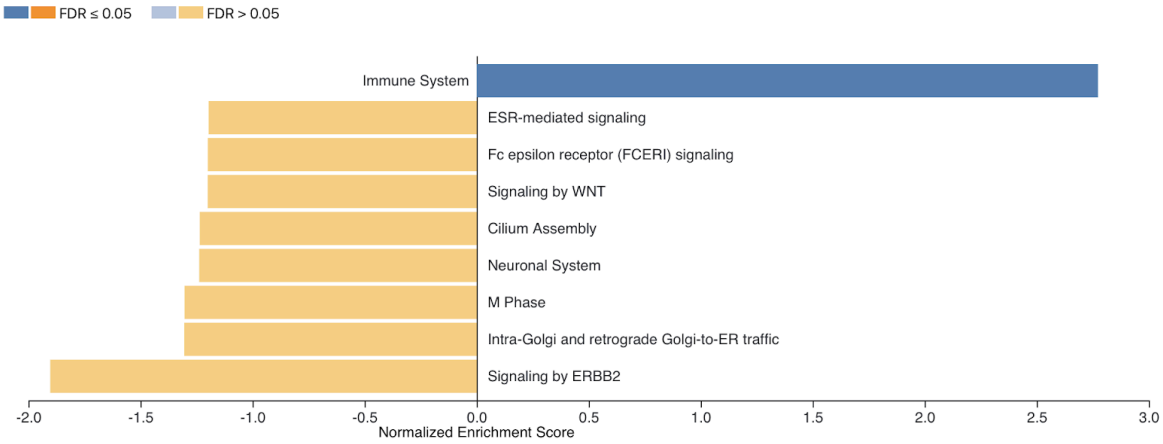
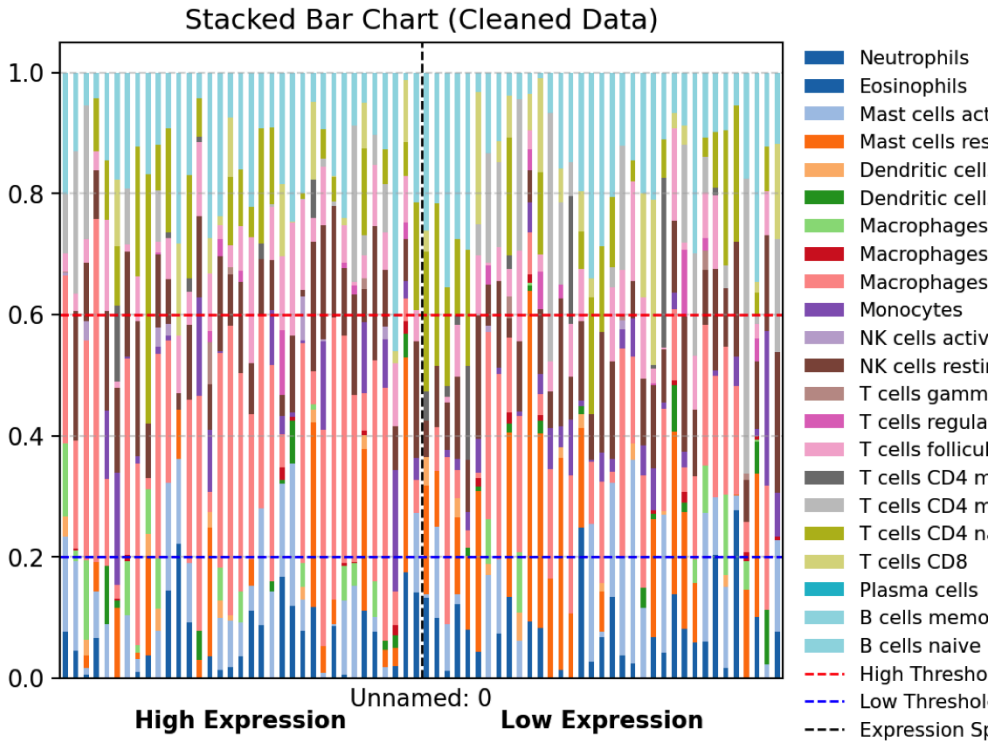
Weighted set cover



Enrichment Categories: pathway_Reactome



Weighted set cover



STACKED BAR PLOT